

Measuring Revenue Management Efficiency

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1 Introduction

In this paper framework for answering the following questions is provided.

1. How to understand that revenue management done well?
2. What is efficiency of revenue management?
3. Is there any reasonable measure of efficiency?
4. How to follow market movements?
5. How to separate market performance and revenue management performance on top of the market?

2 Mathematical models

2.1 Static case

Let $f_1 > \dots > f_n$ be fares for classes. f_1 is the fare of highest class and f_n is the fare of lowest class. Demand is characterized by random vector

$$D = (D_1, \dots, D_n) \sim \mathcal{N}(\mu, \Sigma),$$

where μ is mean vector and Σ is covariance matrix of n -dimensional multivariate normal distribution on some probability space $(\Omega, \mathcal{F}, \lambda)$.

Let

$$P = \{p \in \mathbb{Z}_+^n \mid \sum_{i=1}^n p_i = c\}$$

be a set of all possible allocations among classes from fixed capacity c .

Definition 1 *Mapping*

$$\mathcal{P}_p : \mathbb{R}_+^n \times \mathbb{R}_+^{n \times n} \rightarrow P$$

is called pure strategy.

In other words pure strategy is a function which for each demand vector μ and covariance matrix Σ assigns certain protection limits.

Definition 2 *Mapping*

$$\mathcal{P}_m : \Omega \times \mathbb{R}_+^n \times \mathbb{R}_+^{n \times n} \rightarrow P$$

is called mixed strategy.

The difference between pure and mixed strategy is that mixed strategy for μ and Σ assigns protection limits with some probabilities. Apparently any pure strategy is a mixed strategy.

Definition 3 *Blind strategy is a mixed strategy which is constant in μ and Σ and independent of D .*

Revenue under some mixed strategy $\mathcal{P}_m(\omega) = (p_1(\omega), \dots, p_n(\omega))$ is given by

$$R(\mathcal{P}_m) = \sum_{i=1}^n f_i \cdot \min\{D_i(\omega), p_i(\omega)\}$$

and is, indeed, random variable with some distribution.

Consider also

$$\hat{R}(\mathcal{P}_m) = \sum_{i=1}^n f_i \cdot \min\{\mu_i, p_i(\omega)\}.$$

When demand gets deterministic, i.e. Σ is identity matrix then

$$R(\mathcal{P}_m) = \hat{R}(\mathcal{P}_m).$$

Definition 4 *Blind performance is*

$$BP = E[R(\mathcal{P}_b)]$$

Definition 5 *Blind volatility is*

$$BV = \sqrt{\text{Var}(R(\mathcal{P}_b))}$$

Definition 6 *Market volatility is*

$$MV = BV - \sqrt{\text{Var}(\hat{R}(\mathcal{P}_b))}$$

Suppose actual realized revenue is

$$R = \sum_{i=1}^n f_i \cdot b_i.$$

Definition 7 *Efficiency is*

$$\epsilon = \frac{R - BP}{BV}.$$

2.2 Dynamic case

3 Applications

3.1 Revenue Mangement Efficiency

3.2 Test for skills

3.3 Risk awarness and VaR

4 Implementation

To calculate ϵ we need to find distribution of random variable $R(\mathcal{P}_b)$. Potentially given certain blind strategy one can find closed-form of probabily density function of $R(\mathcal{P}_b)$.

In general much easier approach is to simulate random variable $R(\mathcal{P}_b)$ and consider sample mean and sample variance as estimates for BP and BV .

If n is the number of booking classes and m is the number of simulation runs then complexity of algorithm is $O(nm)$.

5 Discussion

Approach is taken above to define revenue management efficiency is borrowed from portfolio theory. Random portfolios are used there to analyze and give score to investment funds or particular portfolio managers. Idea is simple knowing past history of the market one could ask what if portfolio was managed in completely random fashion and not utilizing any knowledge of the market. Random fashion means here that at each point of time decisions to sell or to buy were done independently of market signals. Simulating this random behavior using past history yields the value of the portfolio on current date. This simulated value is compared then with actual realized value of portfolio and the gap between these two values gives an idea of performance of certain mutual fund or manager.

One of advantages of this method is that it allows to separate market changes and performance of manager. Value of random portfolio incorporates information on the market change and serves as reliable benchmark to measure the performance. Traditional benchmarks as compare-to-last-year are lacking this property.

Application of idea from portfolio theory to revenue management is straightforward. Suppose when flight departed we are given forecast of demand for each class and actual bookings in each class. Question we want to answer is whether the flight was managed well. In the same way as in portfolio theory we define random fashion of managing the flights which is the process of assigning protection limits not utilizing any knowledge of forecast. Simulating this random behavior of revenue manager brings some average revenue which is achieved in current market conditions when no revenue management is applied. The difference between actual revenue and simulated revenue estimates revenue added by revenue management practice. In order to compare this measure across different markets and time we scale it by variance of simulated revenue.

Some notes on formal definitions. Demand is assumed to follow joint normal distribution, but components could be dependent. This is done to stress that the proposed approach handles both cases dependency and independency easily. Assumption of normality is not restriction either. We made this assumption basically for the ease of introduction further definitions.

Joint normal distribution is fully defined by mean vector and covariance matrix. That is why we define pure strategy as mapping from the set of all possible mean vectors and covariance matrices to the set of all possible class protections. Notice that nesting order of classes is ignored and protection for the class is just the number of bookings we aim to get to this class. We believe that using nesting protection limits will just complicate formulas, but will not lead to different theoretical results.

We define mixed strategy in order to model random behavior in defining protection limits. To our knowledge mixed strategies are not discussed in revenue management literature, but they are well studied and used in general optimiza-

tion theory. The idea behind them is quite simple. Given some knowledge of the market one could artificially randomize his/her behavior. Under certain conditions mixed strategy could do better, but not in classical revenue management settings. Difference of definition compared to pure strategies is just dependency of protection limits on realization of random event ($\omega \in \Omega$).

Blind strategy is model for random behavior not utilizing any knowledge of the market. We define it constant in mean vector and covariance matrix and independent of demand. Fact that it is not utilizing any knowledge of the market is justified by the fact it is the same for different market conditions (mean vector and covariance matrix). Independence of demand and strategy is natural due to realization of demand vector should not anyhow affect the way strategy chosen.

Blind performance is defined naturally through average result achieved with blind strategy. We associate blind performance with market performance. We believe that it is market characteristic and revenue management actions are eliminated through randomization. When it grows it tells about growing market and other way around when it falls it tells us about falling market. Blind volatility characterize variability of random blind result. This incorporates both variability of the market as well as variability of the result due to random management procedure.

Efficiency measure tells how many times measured in the number of standard deviations achieved result is better than result achieved with blind random behavior. We separate blind volatility and market volatility. The reason for this is that blind volatility contains also volatility due to blind strategy. To separate market volatility and blind volatility we define revenue achieved with mixed strategy and deterministic demand (see $\hat{\mathcal{R}}$). We take the difference (?) and call it market volatility.

Advantages:

- Always defined (low and high demand flights).
- Easy to calculate.
- Accounts for statistical properties of demand.
- Could be implemented for any type of demand (normality is not restriction).

Disadvantages:

- Assumes good forecast.
- Understanding of the measure requires familiarity with concept of probability.

6 Experiments

We made simulation of one Finnair flight to BKK for the last year. Picture with graphs is below. On the first graph there is two time series. Green one is actual revenue (PIF multiplied by the number of bookings on departure date) and simulated blind revenue. It is seen that actual revenue is always above of simulated revenue, but there is some periods when the distance between actual

and simulated one is quite small. This indicates that in these periods revenue management could be improved. Good example of this time period is period just before Christmas last year. Simulated revenue is very close to actual one. On other hand there are also positive examples when for falling down market revenue management was adjusted such that efficiency stayed on the same level.

Second graph plots defined above efficiency measure and the last one plots market volatility defined above as well.

References

- [1] Sumkin is kot.

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